

# Oliver Fogelin

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Triple citizen (US, UK, Australia) with full work rights in all three countries

## EDUCATION

### University of Cambridge

Cambridge, UK

*BA Computer Science (Hons) – High First, ranked 15th/118*

*Sep. 2023 – June 2026*

- Dissertation: **gvecdb**, a hybrid graph/vector database in OCaml implementing MVCC concurrency control, HNSW indexing from scratch, and AVX-512 SIMD vectorized distance kernels via the OxCaml compiler. Scored 88.5%, ranked 7th in year
- Took modules in **Multicore Semantics and Programming** (lock-free data structures, relaxed memory models, concurrent algorithms), Concurrent and Distributed Systems, Optimising Compilers
- Choral Scholarship

### Churcher's College

Hampshire, UK

*A Levels: A\*A\*A\*A\*A (Further Maths, Maths, Biology, English Literature, AS Computer Science)*

*2020 – 2022*

## EXPERIENCE

### Quantitative Developer

June 2026 – Present

*hLevel LLC*

*New York, NY*

### Software Engineer Intern

June 2025 – Sep. 2025

*Jump Trading*

*London*

- Built firm-wide ClickHouse schema exploration tooling with a web IDE for SQL execution, real-time visualization, and LLM-powered dashboarding. Immediate adoption across dev and trading teams
- Shipped internal libraries in Python, Go, and Typescript used across the firm. Received return offer

### YC Summer Fellowship

June 2025 – Aug. 2025

*Y Combinator (concurrent with Jump internship)*

- Selected as 1 of 29 undergrads internationally from over 10,000 applicants (0.29% acceptance) for a \$20k research grant to build open-source developer tools
- Built stardust: an embeddable hybrid graph-vector database in C++ on LMDB with Cap'n Proto serialization, supporting concurrent graph traversal with approximate nearest-neighbor search

### Software Engineer

Jan. 2025 – June 2025

*Iterate*

*London*

- Built agent infrastructure for early-stage startups at a company backed by OpenAI, Garry Tan, Index Ventures, and Naval Ravikant. Founded by ex-Monzo CTO/Co-Founder
- Architected multi-tenant agent execution platform on Cloudflare Workers for Platforms with sandboxed per-tenant isolation, custom runtime injection, and distributed state management

### Software Engineer Intern

Aug. 2024 – Oct. 2024

*ReactWise (YC S24)*

*San Francisco*

- At 19, sole engineer during the YC S24 batch. Owned entire stack (Python backend, React frontend, infrastructure) from day zero to production deployment at Pfizer in 10 weeks
- Built full-stack app for Bayesian optimisation of chemical reactions, integrating with pharma high-throughput experimentation workflows and optimising transfer learning in deployment

### Founding Engineer

Nov. 2022 – Sep. 2023

*Worldover (Entrepreneur First, Index Ventures)*

*London*

- First engineer hire at an Index Ventures-backed graph intelligence startup. Grew and led the engineering team to 7. Owned Python backend, data model design (AWS Neptune/DynamoDB for highly relational entity graphs), and the full NextJS frontend
- Designed and deployed cloud infrastructure from scratch (API Gateway, Lambda, CloudFormation). Left to start Cambridge

- Founded a software consultancy at 14. Built and shipped production software for the British Physics Olympiad, AppyWay (TfL-partnered parking data), the Town of Littleton NH, and dozens of other clients over 3.5 years
- Projects ranged from full-stack web apps to embedded systems, including reverse-engineering an LED sign's firmware over Wireshark with only Mandarin documentation

## PROJECTS

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- YENDOR** | *C, Bison, Emscripten/WASM, OpenGL* | [github](#) | [play](#) Dec. 2025
- **Won 1st place overall** in the Langjam Gamejam (34 entries, run by Austin Henley, CMU professor)
  - Designed **nh**, a novel programming language compiled via Bison to C, then to WASM via Emscripten. Built a NetHack-inspired roguelike entirely in nh with a custom garbage-collected C runtime and OpenGL bindings, all in 7 days
- gvecdb** | *OCaml, OCaml, AVX-512, mmap* | Dissertation 2026
- Hybrid graph/vector database implementing MVCC concurrency control from scratch, HNSW approximate nearest-neighbor indexing, and AVX-512 SIMD vectorized distance computations via the OCaml unboxed types compiler extension
  - Memory-mapped storage engine with crash-consistent B-tree indexes and custom memory allocator. Benchmarked against FAISS and pgvector
- stardust** | *C++, LMDB, Cap'n Proto* | [github](#) Aug. 2025
- Embeddable hybrid graph-vector database supporting concurrent HNSW-based ANN search integrated with labeled property graph traversals. Zero-copy serialization via Cap'n Proto, LMDB-backed persistence
- tripos.pro** | *Typescript, Python, RAG* | [live](#) Nov. 2024
- Past paper tracker with semantic search over Cambridge CS exam content. Used by over half the current Computer Science cohort (~200 students). Clean UI with efficient search and product features driven by user feedback
- eotu** | *Typescript, LLM integration* | [play](#) July 2024
- Idle game where an LLM is the player's sole interface. Natural language commands drive a sci-fi resource management game with emergent gameplay. Prompt engineering as game design
- SINAESTHESIA** | *Godot, GDScript* | [play](#) Feb. 2025
- Synesthetic arcade game where each sense (sight, sound, touch) maps to a different combat mechanic. Built in Godot with custom shader pipelines for sensory feedback systems
- musejump** | *Go, Typescript, Spotify API* | [github](#) Apr. 2025
- Tool for setting up skips and jumps within Spotify tracks. Go backend with real-time Spotify playback control via OAuth, React frontend for waveform editing